Subjective Wellness, Acute:Chronic Workloads, and Injury Risk in College Football

John A. Sampson,[1] Andrew Murray,[2] Sean Williams,[3] Adam Sullivan,[4] and Hugh H. K. Fullagar[5]

[1]*Center for Human and Applied Physiology, School of Medicine, University of Wollongong, Wollongong, New South Wales, Australia;* [2]*Department of Athletics, University of Oregon, Eugene, Oregon;* [3]*Department for Health, University of Bath, Bath, United Kingdom;* [4]*Research Institute for Sport and Exercise Sciences, Liverpool John Moores University, Liverpool, United Kingdom; and* [5]*Sport and Exercise Discipline Group, University of Technology Sydney, Lindfield, New South Wales, Australia*

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Abstract

Sampson, JA, Murray, A, Williams, S, Sullivan, A, and Fullagar, HHK. Subjective wellness, acute:chronic workloads and injury risk in college football. *J Strength Cond Res* XX(X): 000–000, 2018—Acute:chronic workload ratios (ACWRs) are associated with injury risk across team sports. In this study, one season of workload and wellness data from 42 collegiate football players were retrospectively analyzed. Daily 7:21 day exponentially weighted moving average (EWMA) ACWRs were calculated, and *z*-score fluctuations ("normal," "better," and "worse") in sleep, soreness, energy, and overall wellness were assessed relative to the previous day ACWRs and considered as an interactive effect on the risk of noncontact injury within 0–3 days. Fifty-five noncontact injuries were observed, and injury risks were *very likely* higher when ACWRs were 2 *SD*s above (relative risk [RR]: 3.05, 90% confidence interval [CI]: 1.14–8.16) and below (RR: 2.49, 90% CI: 1.11–5.58) the mean. A high ACWR was *trivially* associated ($p < 0.05$) with "*worse*" wellness ($r = -0.06$, CI: -0.10 to -0.02), muscle soreness ($r = -0.07$, CI: -0.11 to -0.03), and energy ($r = -0.05$, CI: -0.09 to -0.01). Feelings of *"better"* overall wellness and muscle soreness with collectively high EWMA ACWRs displayed *likely* higher injury risks compared with "*normal*" (RR: 1.52, 90% CI: 0.91 to 2.54; RR: 1.64, 90% CI: 1.10–2.47) and *likely* or *very likely* (RR: 2.36, 90% CI: 0.83 to 674; RR: 2.78, 90% CI: 1.21–6.38) compared with "*worse*" wellness and soreness, respectively. High EWMA ACWRs increased injury risk and negatively impacted wellness. However, athletes reporting "better" wellness, driven by "better" muscle soreness presented with the highest injury risk when high EWMA ACWRs were observed. This suggests that practitioners are responsive to, and/or athletes are able to self-modulate workload activities.

Address correspondence to John A. Sampson, jsampson@uow.edu.au.

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Introduction

American football is a physically demanding contact sport comprising substantial impact loads and intermittent bouts of high-intensity activity (45,46). Injury rates are correspondingly high and likely associated with the heavy contact loads; however, >25% of injuries are attributed to preventable noncontact injury (8). In college football, athletes are typically engaged in 8–9 $h \cdot d^{-1}$ of football-related activities in addition to 3–4 $h \cdot d^{-1}$ in academic classes and home study. The varied injury risks observed across positional groups and with playing experience (relative to educational enrollment status) may yet be a consequence of diverse training and game demands (30). Monitoring, modifying, and optimizing workloads in college football in an attempt to reduce the number of these injuries is thus an essential player welfare practice (10).

Workload monitoring is indeed commonplace, with global positioning systems (GPSs) and built in inertial measurement units (IMUs) typically used in college football to quantify training and match workloads (37,45–47). Across a range of contact team sports, including American college football, increased injury risks have consistently been observed when "spikes" in current (acute) relative to accumulated (chronic) GPS-/IMU-derived acute:chronic workload ratios (ACWRs) are observed (7,19,37). The consistency of increased injury risk seen across the literature when high ACWRs occur suggests the ratio has merit for workload monitoring practice. However, where absolute (%) risks are reported, ≤25% of athletes exposed to high and very high ACWR actually suffer an injury (19), and low-predictive capabilities have been observed (9,29). In this regard, one should consider that many sports encompass a range of external training stressors (e.g., running, throwing, contact, resistance training, and static work) that contribute to the total workload, and it is important to recognize that

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increased injury risks do not arise from workload spikes per se, but from the stress associated with threats to homeostasis by separate and potentially multiplicative intrinsic and extrinsic disturbances (5). Correspondingly, it has been shown that athletes possessing greater fitness are less likely to sustain injury when exposed to ACWR spikes and recover more rapidly from competition-induced workloads (20,25,27). Indeed, in American College football, although workload "spikes" are informative, some athletes are shown to be more robust and less susceptible to injury when workload spikes are observed (37).

A number of current studies have examined the multiplicative effects of combing external workload measures with consistently greater risks observed with low chronic workloads and a concurrently high ACWR (7,37). Notably, Colby et al. report substantially increased injury risks with heavy nonsport activity and old lower-limb pain (7). Pain is commonly reported among athletes and may reflect microtrauma associated with overuse injury (6). Considering the high prevalence of overuse injury (15), and reports of athletes frequently participating despite the presence of pain (36,42), methods for monitoring player wellness are well justified. Indeed, subjective internal stress reports including soreness, sleep, stress, and fatigue have been shown to reflect negative responses to high training loads and the frequency of high-intensity activity and collisions in sport (33,40,43). However, we are unaware of any research that has assessed the effect of external workload "spikes" depicted by ACWR on an athlete's subsequent internal self-reported wellness.

Considering quantitative data depicting the athletes' internal stress response from wellness reports alongside fluctuating workloads in sport may also provide further insight into an athlete's risk of injury. The current investigation will therefore assess the effect of fluctuating ACWRs on self-reported wellness and examine ACWR-wellness interactions relative to the risk of injury in NCAA American college football.

Methods

Experimental Approach to the Problem

Athletic workload and self-reported (subjective) wellness questionnaires collated over a full season (17 weeks) of NCAA Division 1 college football were retrospectively analyzed. Previously, a 7:21 day coupled ACWR calculated using an exponentially weighted moving average (EWMA) method with a 3-day injury lag period has shown the greatest associations with injury (37). In this article, 7:21 day EWMA ACWR were synchronized with wellness data reported the morning after 3 × weekly main field-training sessions. Any daily file missing self-reported wellness data were removed leaving 1,807 aligned wellness/ACWR in-season data files (training days) in the analysis.

Subjects

Forty-two athletes competing for the same Division I-A American college football team (mean ± *SD*: age: 20.5 ± 1.2 years, mass: 102.8 ± 17.4 kg, and height: 186.4 ± 6.7 cm) comprising 7 defensive backs, 8 defensive linemen, 6 linebackers, 8 offensive linemen, 2 quarterbacks, 5 running backs, 5 wide receivers, and 1 tight end were included in this study. Within this group, 7 were freshman, 7 juniors, 12 sophmores, and 16 were seniors. All participants signed an informed consent form on enrollment indicating that de-identified data collected as part of their athletic participation may be used for research purposes. Participants were specifically informed of the requirements of this study before data collection, and all experimental procedures were approved by the University of Wollongong human ethics committee and University of Oregon Research Compliance Services.

Procedures

Injuries. Injuries were recorded and documented by the team's athletic training group and classified by incident; date; location; type; and mechanism. As per previous research, diagnoses made by athletic training staff were reviewed retrospectively and confirmed or amended by a sports physician (30). All noncontact injuries reported to medical staff in this investigation resulted in some form of withdrawal from practice or game time, and all were included in the analysis (regardless of ensuing time-lost or not on subsequent days) because this type of injury is considered largely preventable (12).

Quantifying Load. Workloads were collected from GPS sampling at 10 Hz (Optimeye S5; Catapult Innovations, Melbourne, Australia) during the 3-week preseason conditioning phase, all in-season "on-field" workloads (comprising 3 × weekly conditioning sessions and 2 × weekly walk-through sessions) and game day. Data collected by this device are considered a valid and reliable reflection of the activities performed in team sports (21,41). Only players with workload data from every type of session (preseason conditioning, in-season conditioning, and walk-through days) were included in the analysis. This decision was made to include a value for any "missing" data files (typically due to a malfunctioning GPS unit) in the data. In this article, 37 "missing" preseason (generalized conditioning) files were included relative to the players' individual weekly preseason average. During the in-season, the individuals' average specific to the missing session (GPS devices were typically only worn during 1 of the 2 weekly walk-through sessions and for 60 missing conditioning sessions) were added to the data set. Participants wore the same GPS unit in each session. Playerload, a variable collected by triaxial accelerometers within the device sampling at 100 Hz and calculated within the manufacturer's software as the square root of the sum of the squared instantaneous rate of change in acceleration within the 3 planes divided by 100 (OpenField 1.11; Catapult Innovations), was used to quantify workload. Daily EWMA ACWRs were retrospectively calculated by dividing the 7-day (acute), by the 21-day (chronic) workload (37).

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Subjective Wellness. Each day's EWMA ACWR was aligned with wellness reported in a customized wellness questionnaire ~2 hours before each field-training session (11). No data were collected on, or the day after game day (rest day/day off). The questionnaire comprised three 5-point Likert scale questions on self-reported soreness (1 = terribly sore, to 5 = no soreness at all), sleep (1 = slept terrible, to 5 = excellent sleep), and energy (1 = no energy, to 5 = totally energized), and participants were familiarized with all scales. Overall wellness was calculated as the average of the summed soreness, sleep, and energy scores for each athlete (1 = poor wellness, to 5 = excellent wellness).

Data Analysis. *Z*-score deviations relative to an individual's own mean or "*normal*" score were calculated and expressed as "*better*" (≥1 higher than the mean) or "*worse*" (≤1 lower than the mean) to determine a meaningful change in wellness, sleep, soreness, and energy. The daily ACWRs were aligned with the associated self-reported wellness scores (e.g., calculated ACWRs after Monday's session were aligned with self-reported wellness z scores recorded on Tuesday morning) providing 3 ACWR/wellness data points per week.

Statistical Analyses

All estimations were made using the *lme4* package (1) with *R* (version 3.3.1; R Foundation for Statistical Computing, Vienna, Austria). The subjective wellness reports were assessed for normality and appropriate parametric or nonparametric correlations performed. A generalized linear mixed-effect model (GLMM) with the complementary log-log link function was used to model the association between ACWR, wellness measures, and injury risk in the subsequent 3-day period. Acute:chronic workload ratios and wellness measures were modeled as fixed-effect predictor variables, and player identity was the random effect. A multiplicative term was included in the model to assess the interaction between ACWR and wellness measures. The odds ratios obtained from the GLMM model were converted to relative risks (RRs) to interpret their magnitude (18). The smallest important increase in injury risk was a RR of 1.11, and the smallest important decrease in risk was 0.90 (17). An effect was deemed "*unclear*" if the chance that the true value was beneficial was >25%, with odds of benefit relative to odds of harm (odds ratio) of <66. Otherwise, the effect was deemed clear and was qualified with a probabilistic term using the following scale: <0.5%, *most unlikely*; 0.5–5%, *very unlikely*; 5–25%, *unlikely*; 25–75%, *possible*; 75–95%, *likely*; 95–99.5%, *very likely*; and >99.5%, *most likely*. The data are presented as mean values and 90% confidence intervals (CIs) with injury likelihoods estimated at typically very low (−2 *SD*), low (−1 *SD*), mean, high (+1 *SD*), and very high (+2 *SD*) values of ACWR. These values were equivalent to ACWRs of 0.44, 0.67, 0.91, 1.14, and 1.38, respectively.

Results

A total of 55 noncontact injuries were observed in this data set with 27 occurring in game time, 2 during strength-based conditioning, and 26 during field-based practice sessions. Forty-two injuries were reported in the lower body affecting the ankle (15), knee (11), foot (5), posterior thigh (5), hip (5), and toe (1). The remaining 13 injuries were observed at the lumbar spine and lower back (7), shoulder (5), and elbow (2). A sprain or strain of the affected area encompassed 67% of all injuries and the outstanding 33% comprised 3 or less diagnosed cases of bursitis, herniated disc, generalized pain, tendinitis, subluxation, plantar fasciitis, patellofemoral disorder, muscular imbalance, impingement, cyst, hyperextension, or dysfunction.

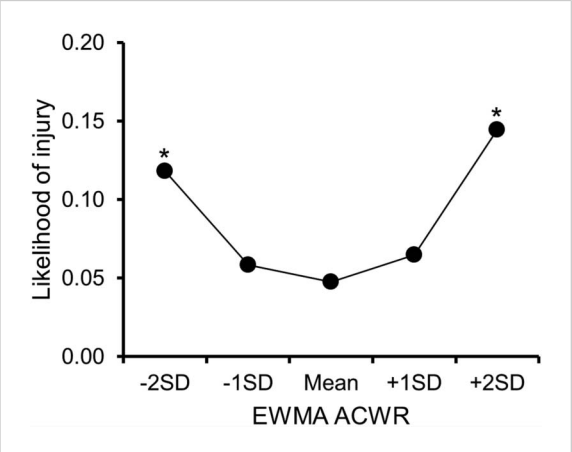

Figure 1. Predicted probability of injury in college football players with deviations from the mean EWMA ACWRs. ACWR = acute:chronic workload ratio. * = *very likely* higher injury risk compared to mean.

Injury Risk and Daily Acute:Chronic Workloads

The mean ACWR observed in this study was 0.91 ± 0.23. A characteristic rise in the probability for injury was observed

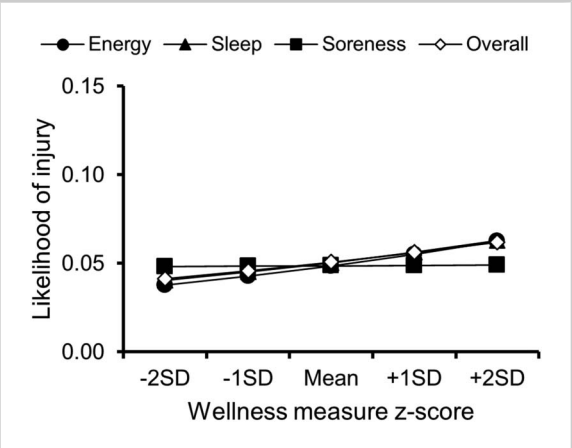

Figure 2. Predicted probability of injury in college football players with deviations from the mean subjectively reported sleep, soreness, energy, and overall wellness.

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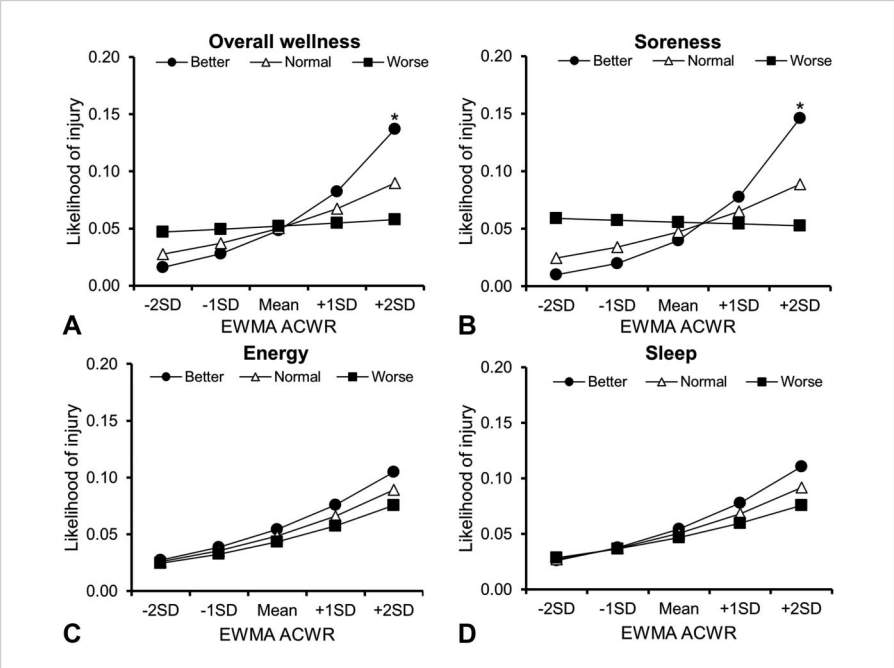

Figure 3. Interactive effect of a deviation from the mean EWMA ACWRs when collectively considering an athlete's state of perceived (A) overall wellness, (B) soreness, (C) energy, and (D) sleep quality. ACWR = acute:chronic workload ratio.

with high and low ACWRs (Figure 1). Specifically, injury risks were *very likely* higher when the ACWR was 2 *SD*s above the mean (RR: 3.05, 90% CI: 1.14–8.16) and 2 *SD*s below the mean (RR: 2.49, 90% CI: 1.11–5.58), when compared with the mean ACWR.

Injury Risk and Wellness

Across the data set, typical mean wellness 3.23 ± 0.65, sleep 3.32 ± 0.83, energy 3.34 ± 0.78, and soreness 3.05 ± 0.88 were reported. No clear effect on the likelihood of injury with "*better*" (>+1 *SD*) or "*worse*" (<−1 *SD*) reports of wellness, sleep, energy, or soreness was observed (Figure 2).

Effect of Acute:Chronic Workload Ratio on Wellness

Normality across the data set was not observed for any wellness variable, and Spearman's correlations between the previous days' EWMA ACWR with sleep, energy, soreness, and overall wellness were performed. Significant ($p < 0.05$), although *trivial* associations were observed when examining the change (Z score) in subjective ratings with "*worse*" scores in overall wellness ($r = -0.06$, CI −0.10 to −0.02), muscle soreness ($r = -0.07$, CI −0.11 to −0.03), and energy ($r = -0.05$, CI −0.09 to −0.01) observed when a higher ACWR was recorded the previous day.

Wellness, Acute:Chronic Workloads Interactions, and Injury Risk

Acute:chronic workload ratio and wellness interactions highlight that individuals subjectively reporting "*better*" wellness when exposed to a high (+2 *SD*) ACWR had a *likely* higher risk of injury in the subsequent 3 days compared with those reporting "*normal*" (RR: 1.52, 90% CI: 0.91–2.54) or "*worse*" levels of wellness (RR: 2.36, 90% CI: 0.83–6.74) (Figure 3). No clear interactions were observed when examining subjective sleep ($p = 0.74$) or energy ($p = 0.88$) and ACWR associations with injury. However, a *likely* and *very likely* increase in the probability of injury was observed when high ACWR (+2 *SD*) and "*better*" muscle soreness were collectively observed in comparison with "*normal*" (RR: 1.64, 90% CI: 1.10–2.47) and "*worse*" soreness levels (RR: 2.78, 90% CI: 1.21–6.38) (Figure 3).

Discussion

In this investigation of collegiate American Football, low and high ACWRs increased the risk of injury. Our results highlight subsequently lower wellness, energy, and increased muscle soreness after days that evoked high EWMA ACWRs. Interestingly, however, the greatest risk of sustaining an injury (within 3 days) was observed when high ACWR and typically "*better*" perceived wellness, driven by perceived levels of soreness, were collectively observed. To our knowledge, this study is the first to assess the relationship between an athlete's ACWR and their state of wellness the following day, and the first to consider interactions between the ACWR and perceived wellness relative to the risk of injury.

Playerload was the chosen workload measure given its suitability for encompassing both indoor and outdoor training comprising acceleration, deceleration, sprint, and contact efforts (4,34) and the frequency of these activities in college football (45,46). Increased injury risks were observed at lower ACWRs than those commonly reported; however, the characteristic "U" curve depicting a "sweet spot" at moderate ACWR and injury risks 2.5 to 3 times greater with lower and higher ratios (13) was apparent. In practical terms, the change in workload associated with higher rates of injury at each end of the spectrum represented a relative increase or decrease in load of >40–50% that is consistent with ACWR-injury risks observed across a larger cohort of this group (37). High-risk scenarios that may result in the high ACWR and lead to injury in college football such as "return to play" and unaccustomed game time have been proposed

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(37). However, despite the very likely higher injury risks associated with fluctuations of ±2 *SD* from the mean workload in this cohort, the absolute risk did not exceed 15%. Considering the negative effect of high workloads on an athlete's self-reported wellness (33,40,43), it was anticipated that lower subjective ratings of wellness observed concurrently with high and/or low EWMA ACWRs would amplify injury risks.

No clear associations between any subjective measure of wellness and the likelihood of injury were observed. However, wellness scores indicative of "*worse*" perceived wellness driven by energy and soreness were observed the day after a high ACWR. These associations seem to extend current research by highlighting the impact of workload spikes (generally) on an athlete's internal wellness. Given the deleterious effects that excessive workloads are known to have on an athlete's sleep (16), it was somewhat surprising that no associations with injury and EWMA ACWR workload spikes were observed. However, increased sleep efficiency has previously been observed during intense training in Rugby League players (39), suggesting that the impact of training on sleep may be positive in the absence of an overtrained or functionally overreached status. Nevertheless, given the apparent negative influence of a high ACWR on subjective rating of wellness, and it was anticipated that the risk of injury would correspondingly be amplified with low wellness when considered as multiplicative variables.

It was therefore surprising to observe increased risks were predominantly associated with a high EWMA ACWR when athletes subjectively reported feeling "*better*" driven by perceived levels of soreness. As such, it should first be considered that the negative associations between EWMA ACWR and wellness we observed were *trivial*, and the impact should be interpreted with caution. Furthermore, the association between soreness and high EWMA ACWRs observed in this investigation was likely affected by typically higher workloads on (35) and consistently increased muscle soreness after (11) game day. The impact of games on subjective wellness has also been shown to perpetuate and deteriorate throughout the training week up to 4 days after game (11). Subjective reports of "*worse*" perceptions of wellness before training can reduce training outputs (14,26), and more specifically, "*worse*" muscle soreness has previously been related to a reduction in player effort (Session-RPE) in college football players (15). It is possible that practitioners are responsive to negative wellness perceptions and may have intervened in this investigation to modulate training loads and/or players themselves may have self-regulated reductions in their training effort. Such actions may explain the low sensitivity that ACWR models have shown with injury (9,29). Consistent with this theory, an athlete reporting "*better*" wellness and soreness may alternatively be predisposed to more frequent high intensity activities that are considering injury initiating events such as sprinting, accelerating, and cutting (3,24). Although we acknowledge that this remains speculative, further research focusing on the relationship of daily fluctuations in subjective recovery responses and training outputs is warranted.

The results of the current research do not suggest that adverse wellness increases the risk of injury. The pattern of injury was comparable with those reported in a recent longitudinal study (23), and previous accounts of the daily and seasonal GPS workload distribution in this team (32) are similar to that observed in other groups of NCAA Division I footballers (44). However, a number of limitations must be recognized. First, one should recognize that despite the similarities noted above, the current study is a report of a single season of injuries from a single team. As such, these outcomes may not be consistently reflected across college football when considering the varied training demands/schedules used. Furthermore, although the number of injuries included in this investigation were considered sufficient to detect moderate–strong associations (2), the overall number was relatively low, and the associations observed were likely underpowered by examining interaction effects. Furthermore, in this and many similar investigations examining injury risks and workloads in team sports, only field-based workloads are considered. As such, although wellness may have been impacted on by workloads (such as resistance exercise) that were not measured in this investigation, they were not included in the ACWR calculation. In addition, the variability in workload and injury risk that may be associated with positional demands and experience may have influenced our results (30) and academic, or other non-athletic stressors, which can adversely affect wellness and amplify injury risks (28), were not recorded and could not be considered. Inadvertently, more complex and confounding variables that influence fatigue, wellness, external, and internal stress may thus have contributed to the risk of injury observed (31). The higher injury risk observed with high workloads and "better" wellness observed in this study may suggest that these confounding variables did not influence our results. However, the accuracy of the wellness reports used in this investigation should also be considered. Variations in wellness relative to game day have previously been observed from the 5-point Likert scale used in this investigation (11); the assessment thus seems sensitive to workloads inducing fatigue. At present, the REST-Q is, however, the only wellness questionnaire that seems to have empirical evidence to show reliability relative to acute and chronic load variations (38).

In this investigation, athletic workload spikes resulted in reduced perceptions of wellness the following day; however, the relationship was trivial. By contrast, the most at-risk group was athletes reporting "*better*" wellness driven by energy and muscle soreness. We suggest that this unexpected association may be a consequence of responsive practitioners applying interventions when negative perceptions of wellness are observed and/or effective self-modulation from players themselves. In this regard, it is also possible

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that high-intensity activities, which evoke an inherently greater risk of injury, occur more frequently when athletes report "*better*" wellness. Future studies examining acute injury risks relative to wellness and high-intensity activities are thus warranted.

Practical Applications

Collectively, this study supports the use of simple non-invasive wellness measures to complement, injury monitoring, and external load constructs within an effective athlete monitoring system for American Football. Specifically, we suggest practitioners (a) apply wellness monitoring within their daily practice to understand the affect and effect of training workloads; (b) where possible, utilize an EWMA ACWR and avoid daily fluctuations >1 *SD* of a player's average; and (c) closely monitor the workload and its composition relative to the planned activity, avoiding unplanned increases in workload even if "*better*" wellness is apparent.

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References

1. Available at: http://cran.r-project.org/web/packages/lme4/lme4.pdf. Accessed August 1, 2014.
2. Bahr, R and Holme, I. Risk factors for sports injuries–a methodological approach. *Br J Sport Med* 37: 384–392, 2003.
3. Bahr, R and Krosshaug, T. Understanding injury mechanisms: A key component of preventing injuries in sport. *Br J Sport Med* 39: 324–329, 2005.
4. Barreira, P, Robinson, MA, Drust, B, Nedergaard, N, Raja Azidin, RMF, and Vanrenterghem, J. Mechanical player load using trunk-mounted accelerometry in football: Is it a reliable, task-and player-specific observation? *J Sport Sci* 35: 1674–1681, 2016.
5. Chrousos, GP and Gold, PW. The concepts of stress and stress system disorders: Overview of physical and behavioral homeostasis. *JAMA* 267: 1244–1252, 1992.
6. Clarsen, B, Rønsen, O, Myklebust, G, Flørenes, TW, and Bahr, R. The Oslo Sports Trauma Research Center questionnaire on health problems: A new approach to prospective monitoring of illness and injury in elite athletes. *Br J Sport Med* 48: 754–760, 2014.
7. Colby, MJ, Dawson, B, Peeling, P, Heasman, J, Rogalski, B, Drew, MK, et al. Multivariate modelling of subjective and objective monitoring data improve the detection of non-contact injury risk in elite Australian footballers. *J Sci Med Sport* 20: 1068–1074, 2017.
8. Dick, R, Ferrara, MS, Agel, J, Courson, R, Marshall, SW, Hanley, MJ, et al. Descriptive epidemiology of collegiate men's football injuries: National Collegiate Athletic Association Injury Surveillance System, 1988–1989 through 2003–2004. *J Athl Train* 42: 221–233, 2007.
9. Fanchini, M, Rampinini, E, Riggio, M, Coutts, AJ, Pecci, C, and McCall, A. Despite association, the acute: Chronic work load ratio does not predict non-contact injury in elite footballers. *Sci and Med Football* 2: 108–114, 2018.
10. Fullagar, HH, McCunn, R, and Murray, A. Updated review of the applied physiology of American college football: Physical demands, strength and conditioning, nutrition, and injury characteristics of America's favorite game. *Int J Sports Physiol Perform* 12: 1396–1403, 2017.
11. Fullagar, HHK, Govus, A, Hanisch, J, and Murray, A. The time Course of perceptual recovery markers following match play in Division I-a collegiate American footballers. *Int J Sports Physiol Perform* 12: 1264–1266, 2017.
12. Gabbett, TJ. The development and application of an injury prediction model for noncontact, soft-tissue injuries in elite collision sport athletes. *J Strength Cond Res* 24: 2593–2603, 2010.
13. Gabbett, TJ. The training–injury prevention paradox: Should athletes be training smarter and harder? *Br J Sport Med* 50: 273–280, 2016.
14. Gallo, TF, Cormack, SJ, Gabbett, TJ, and Lorenzen, CH. Pre-training perceived wellness impacts training output in Australian football players. *J Sport Sci* 34: 1445–1451, 2016.
15. Govus, AD, Coutts, A, Duffield, R, Murray, A, and Fullagar, H. Relationship between pre-training subjective wellness measures, player load and rating of perceived exertion training load in American college football. *Int J Sports Physiol Perform* 13: 95–101, 2018.
16. Hopkins, WG. A spreadsheet for deriving a confidence interval, mechanistic inference and clinical inference from a p value. *Sport Sci* 11: 16–20, 2007.
17. Hopkins, WG. Linear models and effect magnitudes for research, clinical and practical applications. *Sportscience* 14: 49–57, 2010.
18. Hopkins, WG, Marshall, SW, Quarrie, KL, and Hume, PA. Risk factors and risk statistics for sports injuries. *Clin J Sport Med* 17: 208–210, 2007.
19. Hulin, BT, Gabbett, TJ, Lawson, DW, Caputi, P, and Sampson, JA. The acute:chronic workload ratio predicts injury: High chronic workload may decrease injury risk in elite rugby league players. *Br J Sport Med* 50: 231–236, 2015.
20. Johnston, RD, Gabbett, TJ, Jenkins, DG, and Hulin, BT. Influence of physical qualities on post-match fatigue in rugby league players. *J Sci Med Sport* 18: 209–213, 2015.
21. Johnston, RJ, Watsford, ML, Kelly, SJ, Pine, MJ, and Spurrs, RW. Validity and interunit reliability of 10 Hz and 15 Hz GPS units for assessing athlete movement demands. *J Strength Cond Res* 28: 1649–1655, 2014.
22. Jürimäe, J, Mäestu, J, Purge, P, and Jürimäe, T. Changes in stress and recovery after heavy training in rowers. *J Sci Med Sport* 7: 335–339, 2004.
23. Krill, MK, Borchers, JR, Hoffman, JT, Krill, ML, and Hewett, TE. Analysis of football injuries by Position Group in Division I College Football: A 5-year program review. *Clin J Sport Med* 2018. Epub ahead of print.
24. Krosshaug, T, Andersen, TE, Olsen, OO, Myklebust, G, and Bahr, R. Research approaches to describe the mechanisms of injuries in sport: Limitations and possibilities. *Br J Sport Med* 39: 330–339, 2005.
25. Malone, S, Owen, A, Mendes, B, Hughes, B, Collins, K, and Gabbett, TJ. High-speed running and sprinting as an injury risk factor in soccer: Can well-developed physical qualities reduce the risk? *J Sci Med Sport* 21: 257–262, 2018.
26. Malone, S, Owen, A, Newton, M, Mendes, B, Tiernan, L, Hughes, B, et al. Wellbeing perception and the impact on external training output among elite soccer players. *J Sci Med Sport* 21: 29–34, 2018.
27. Malone, S, Roe, M, Doran, DA, Gabbett, TJ, and Collins, KD. Protection against spikes in workload with aerobic fitness and playing experience: The role of the acute: Chronic workload ratio on injury risk in elite Gaelic football. *Int J Sports Physiol Perform* 12: 393–401, 2017.
28. Mann, JB, Bryant, KR, Johnstone, B, Ivey, PA, and Sayers, SP. Effect of physical and academic stress on illness and injury in Division 1 college football players. *J Strength Cond Res* 30: 20–25, 2016.
29. McCall, A, Dupont, G, and Ekstrand, J. Internal workload and non-contact injury: a one-season study of five teams from the UEFA Elite Club Injury Study. *Br J Sport Med* 52: 1517–1522, 2018.

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30. McCunn, R, Fullagar, HH, Williams, S, Halseth, TJ, Sampson, JA, and Murray, A. Playing experience and position influence injury risk among NCAA Division I collegiate footballers. *Int J Sports Physiol Perform* 10: 1297–1304, 2017.
31. Meeuwisse, WH. Athletic injury etiology: Distinguishing between interaction and confounding. *Clin J Sport Med* 4: 171–175, 1994.
32. Murray, A, Fullagar, HH, Delaney, JA, and Sampson, J. Bradford factor and seasonal injury risk in Division IA Collegiate American Footballers. *Sci Med Football* 2: 173–176, 2018.
33. Roe, G, Darrall-Jones, J, Till, K, Phibbs, P, Read, D, Weakley, J, et al. The effect of physical contact on changes in fatigue markers following rugby union field-based training. *Eur J Sport Sci* 17: 647–655, 2017.
34. Roe, G, Halkier, M, Beggs, C, Till, K, and Jones, B. The use of accelerometers to quantify collisions and running demands of rugby union match-play. *Int J Perform Anal Sport* 16: 590–601, 2016.
35. Rogalski, B, Dawson, B, Heasman, J, and Gabbett, TJ. Training and game loads and injury risk in elite Australian footballers. *J Sci Med Sport* 16: 499–503, 2013.
36. Roos, KG, Marshall, SW, Kerr, ZY, Golightly, YM, Kucera, KL, Myers, JB, et al. Epidemiology of overuse injuries in collegiate and high school athletics in the United States. *Am J Sports Med* 43: 1790–1797, 2015.
37. Sampson, J, Murray, A, Williams, S, Halseth, T, Hanisch, J, Golden, G, et al. Injury risk-workload associations in NCAA American College Football. *J Sci Med Sport* 21: 1215–1220, 2018.
38. Saw, AE, Main, LC, and Gastin, PB. Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *Br J Sport Med* 50: 281–291, 2016.
39. Thornton, HR, Duthie, GM, Pitchford, NW, Delaney, JA, Benton, DT, and Dascombe, BJ. Effects of a 2-week high-intensity training Camp on sleep activity of professional rugby league athletes. *Int J Sports Physiol Perform* 12: 928–933, 2017.
40. Thorpe, RT, Strudwick, AJ, Buchheit, M, Atkinson, G, Drust, B, and Gregson, W. Monitoring fatigue during the in-season competitive phase in elite soccer players. *Int J Sports Physiol Perform* 10: 958–964, 2015.
41. Varley, MC, Fairweather, IH, and Aughey, RJ. Validity and reliability of GPS for measuring instantaneous velocity during acceleration, deceleration, and constant motion. *J Sport Sci* 30: 121–127, 2012.
42. Weiss, KJ, McGuigan, MR, Besier, TF, and Whatman, CS. Application of a simple surveillance method for detecting the prevalence and impact of overuse injuries in professional men's basketball. *J Strength Cond Res* 31: 2734–2739, 2017.
43. Wellman, AD, Coad, SC, Flynn, PJ, Climstein, M, and McLellan, CP. Movement demands and perceived wellness associated with preseason training camp in NCAA Division I college football players. *J Strength Cond Res* 31: 2704–2718, 2017.
44. Wellman, AD, Coad, SC, Flynn, PJ, Siam, TK, and McLellan, CP. A comparison of pre-season and in-season practice and game loads in NCAA Division I Football Players. *J Strength Cond Res* 2017. Epub ahead of print.
45. Wellman, AD, Coad, SC, Goulet, GC, and McLellan, CP. Quantification of competitive game demands of NCAA Division I college football players using global positioning systems. *J Strength Cond Res* 30: 11–19, 2016.
46. Wellman, AD, Coad, SC, Goulet, GC, and Mclellan, CP. Quantification of accelerometer derived impacts associated with competitive games in national collegiate athletic association Division I college football players. *J Strength Cond Res* 31: 330–338, 2017.
47. Wilkerson, GB, Gupta, A, Allen, JR, Keith, CM, and Colston, MA. Utilization of practice session average inertial load to quantify college football injury risk. *J Strength Cond Res* 30: 2369–2374, 2016.

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